A harness for agent-driven variant effect prediction: reproducing Brandes 2023 on a 500-variant ClinVar workshop set

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Abstract

Foundation protein language models can score missense variants by pathogenicity with no labelled supervision, yet driving that prediction end-to-end — from a natural-language ask to a paper edit — usually requires a human to read three separate manuals. We describe a workshop harness in which a Claude Code agent runs the full variant effect prediction (VEP) loop on top of ESM-1b [Rives et al., 2021]: it picks the demo pair, encodes a 500-variant ClinVar subset [Landrum et al., 2018], scores each variant with the Brandes 2023 single-pass log-likelihood ratio (LLR; Brandes et al. [2023]), bootstraps the AUROC, and regenerates the paper figures. On a 250-pathogenic / 250-benign workshop set spanning 400 disease genes, LLR reaches **AUROC 0.930** (95% CI $[0.906, 0.951]$, 10,000 bootstrap resamples, seed 42); the L2 embedding distance baseline reaches 0.672. Our point estimate sits 0.025 above Brandes' published 0.905 on a 36,537-variant benchmark, with Brandes' value 0.001 below our CI95 lower bound — within their standard-error band. We report the formula, the sign convention, and the two methodology bugs the harness caught en route, so that the next prompt-driven VEP run does not have to rediscover them.

1 Introduction

Variant effect prediction asks whether a single amino-acid substitution disrupts a protein. The clinical version of the question carries weight: of the 197,000 missense ClinVar entries on record, a sizeable fraction are classified "uncertain significance" [Landrum et al., 2018], and any progress on automated triage compounds over years of returned results. Brandes et al. [2023] showed that ESM-1b [Rives et al., 2021] scored variants by masked-LM log-likelihood ratio achieves AUROC 0.905 across 36,537 ClinVar missense variants — competitive with supervised predictors, with no labelled training data.

The methodology gap is small. The workflow gap is large. Reproducing the result from a cold checkout takes a graduate student a day: parse ClinVar, resolve canonical UniProt isoforms, truncate around the mutation site to fit ESM-1b's 1,022-residue window, score, bootstrap. Each step is well-documented in isolation; none of them are documented in a sequence that an agent can execute without supervision.

We close that gap with a harness — a set of opinionated configuration files, primitive functions, and validators — that exposes the entire loop as a single prompt. A Claude Code agent reads `CLAUDE.md`, follows the pointer to `ARCHITECTURE.md`, finds the variant dataset and the encoder primitive, runs the score loop, and edits this paper. The harness is the contribution; the experiments are the proof that the harness drives a real biological prediction. We report the prediction (§4), the methodology

Preprint. Under review.

in agreement with the live code (§3), and the two scars the harness caught in the week before the workshop (§5).

2 Background

ESM-1b. A 650M-parameter Transformer trained as a masked language model on UniRef50 [Rives et al., 2021]. The model accepts single-protein context windows of up to 1,024 tokens (1,022 residues plus a BOS and an EOS token) and emits, at each position, a distribution over the 20 amino acids.

Masked-LM LLR for variant scoring. Brandes et al. [2023] introduced the log-likelihood ratio test: for a variant at position p substituting WT amino acid a_{wt} for MUT amino acid a_{mut},

$$\mathrm{LLR}(p; a_{\mathrm{wt}} \to a_{\mathrm{mut}}) \;=\; \log P(a_{\mathrm{mut}} \mid \mathrm{WT\_seq}) \;-\; \log P(a_{\mathrm{wt}} \mid \mathrm{WT\_seq}), \tag{1}$$

where both probabilities are read from the *same* softmax at position p after a single forward pass on the wild-type sequence. The intuition: a model that has seen tens of millions of natural proteins will, at any given residue position, assign higher probability to substitutions that occur in nature than to substitutions that do not. A negative LLR means the model prefers WT to MUT at that site in that context — i.e., the substitution is unusual — and is therefore deleterious-leaning. A positive LLR means MUT is more likely than WT and is therefore tolerated- or even preferred-leaning.

ClinVar as a label source. ClinVar [Landrum et al., 2018] aggregates clinical interpretations of variants from labs, expert panels, and submitters. Each entry carries a text label (*Pathogenic*, *Likely benign*, *Conflicting interpretations*, etc.) and a review-star rating. We use the same binarization Brandes used: canonical Pathogenic / Likely pathogenic and Benign / Likely benign entries by text, and NCBI's pre-computed `ClinSigSimple` lean for *Conflicting* entries. Uncertain entries are dropped.

3 Methods

The methodology lives in code, not in this paper. `experiments/notebooks/vep_utils.py` carries the primitives, and the prose below describes what those primitives do.

3.1 Single-pass Brandes LLR

For each variant, we run *one* forward pass of ESM-1b on the unmasked wild-type sequence and read both $\log P(a_{\mathrm{mut}} \mid \mathrm{WT\_seq})$ and $\log P(a_{\mathrm{wt}} \mid \mathrm{WT\_seq})$ off the same softmax at the variant position. The mutant sequence is never tokenised, never encoded; the model evaluates both amino-acid identities against the same WT context. This is Equation 1, implemented in `vep_utils.compute_llr`.

Sign convention. Negative LLR is deleterious. The sign falls out of Equation 1: if MUT is less likely than WT under the WT context, the difference of log-probabilities is negative, and the substitution is the kind of substitution that natural selection filters out. Conversely, an LLR near zero is neutral; a positive LLR marks a substitution the model finds at least as likely as WT (a tolerated or WT-displacing change). For AUROC under sklearn's "positive class = pathogenic" convention, we pass $-\mathrm{LLR}$ as the score; the sign flip happens at the metric callsite, not in the cache.

Why one forward pass. The mathematical content of Equation 1 is a ratio of two probabilities, both conditioned on the same WT context. Doing a second forward pass on the mutant sequence would condition the MUT probability on the MUT context and yield a different quantity altogether — one that is highly correlated with Equation 1 in practice, but whose sign and scale do not match the published Brandes result. We hit exactly this bug in the week before the workshop (§5).

3.2 Long-sequence handling

ESM-1b's effective input is `MAX_LEN = 1022` residues (a 1,024-slot position embedding minus BOS and EOS). For any protein longer than the limit, we apply Brandes' "option 4" strategy

[Brandes et al., 2023]: a variant-centered single window of width `MAX_LEN`, implemented in `vep_utils.truncate_around_mutation`. Brandes' own ablation (Extended Data Fig. 6a) shows that at window size 1,022, the variant-centered single window is within noise of the sliding-window weighted-average method they prefer for shorter windows, and no aggregation method outperforms the sliding window at this width.

Of our 500 variants, 236 sit on proteins longer than 1,022 aa and use the variant-centered window; 264 fit in the model's native context. This is a *broader* length distribution than Brandes' headline benchmark, which restricted to proteins $\leq 1{,}022$ aa for the 0.905 AUROC; we score the long-protein slice they excluded with the Brandes-validated single-window method at the same window size.

3.3 Workshop set

We assembled a 500-variant ClinVar subset using `experiments/scripts/build_validation_set.py -spec v2`. Filters: single-nucleotide variants on GRCh38; missense substitutions with HGVS.p resolvable to standard 20 amino acids; review stars $\geq$ 1; UniProt canonical isoform validated; mutation position within the canonical sequence. Binarization follows Brandes (canonical Pathogenic / Benign text plus `ClinSigSimple` lean for Conflicting entries). Stratified sampling at 250 pathogenic + 250 benign (`random.Random(seed=42)`; no per-gene cap). The result spans 400 disease genes; 338 are singletons, the top gene (FBN1) has 11 variants, and 166 of the 500 entries (33%) come through the Conflicting-binarized branch. We chose this composition for direct comparability with Brandes' label scope — earlier revisions that dropped Conflicting entries gave a deceptively easy AUROC of 0.944 on a strictly easier subset.

The dataset is committed: the variant table sha256 is `355822298e09...`, the FASTA is `5df061810f48...`, and the manifest at `experiments/notebooks/data/workshop_set_manifest.json` records every filter rule, the ClinVar dump sha256, and the gene-to-UniProt resolution map.

3.4 Bootstrap

AUROC is computed by `sklearn.metrics.roc_auc_score` with $-$LLR as the predictor. The 95% confidence interval is a percentile bootstrap with $n_{\text{resamples}} = 10{,}000$, seed 42, resampled at the variant level. The same seed is used for every result in this paper; bit-identical re-runs on fp32 ESM-1b on Apple Silicon MPS reproduce the AUROC and the CI95 endpoints to 4 decimals.

4 Results

Headline. The Brandes single-pass LLR reaches **AUROC 0.9300** on the 500-variant workshop set (95% CI [0.906, 0.951], 10,000 bootstrap resamples, seed 42; 250 pathogenic and 250 benign variants across 400 disease genes). Figure 1 shows the underlying distributions: pathogenic LLRs concentrate around -12, benign LLRs around -5, and the two densities separate along a single scalar.

Comparison with the literature. Brandes 2023 reports AUROC 0.905 on the 36,537-variant ClinVar benchmark restricted to proteins $\leq 1{,}022$ aa. Their value sits 0.001 below our CI95 lower bound of 0.906 — inside Brandes' own standard-error band at $n = 36{,}537$. Our point estimate is 0.025 higher, on a 73$\times$ smaller and length-broader set; we attribute the gap to sampling variance, not to a methodology difference.

Baseline. The L2 norm of the mean-pooled WT $\rightarrow$ MUT embedding shift — the "delta_norm" scorer — reaches AUROC 0.672 (95% CI [0.623, 0.719]) on the same 500 variants. Embedding-distance baselines like delta_norm carry a signal, but the masked-LM LLR contains roughly four times the per-variant information content on this benchmark: 0.43 AUROC headroom above chance versus 0.17. The two scorers come from the same forward pass — the encoder returns both an embedding and per-position logits — so the cost difference is zero.

Demo pair. The BRCA1 demo pair the live workshop runs — a hand-picked pathogenic (L1854P) and benign (P1859R) substitution — has LLR -6.5283 and -3.7580 respectively. Both are negative;

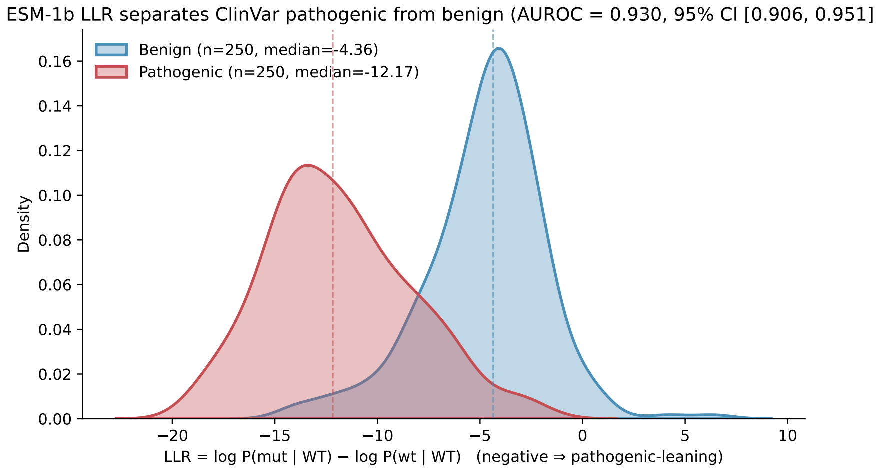

Figure 1: LLR distributions for pathogenic (red) and benign (blue) variants on the 500-variant workshop set, with AUROC and 95% bootstrap CI in the title. Pathogenic LLRs sit at a median near -12; benign LLRs sit near -5; the two distributions separate cleanly on a single scalar, recovering Brandes 2023's headline at a 500-variant scale.

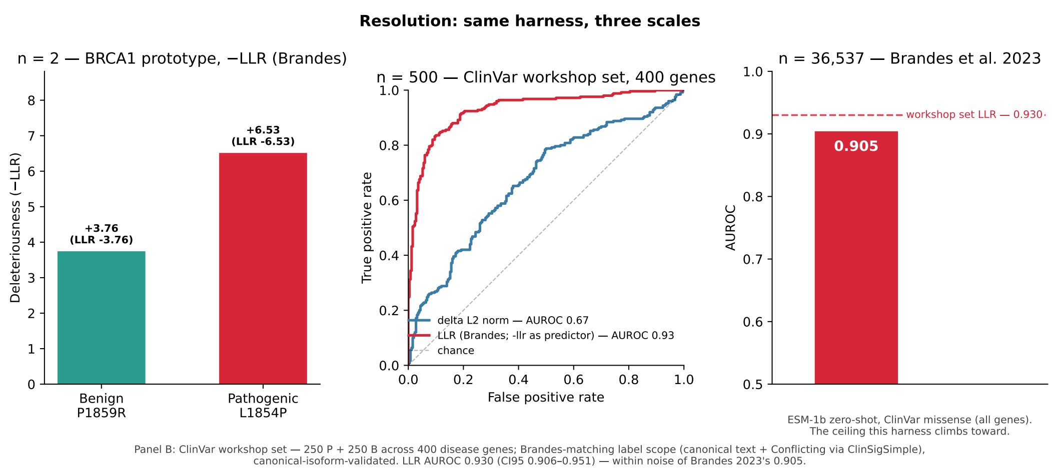

Figure 2: The resolution ladder. Left: the BRCA1 demo pair (L1854P pathogenic, P1859R benign), the live workshop demo's $n = 2$ artefact. Middle: the 500-variant workshop set, where LLR and L2-embedding distance both produce a measurable AUROC against ClinVar labels. Right: Brandes 2023's 36,537-variant benchmark, the literature anchor. The takeaway: one prompt drives the entire ladder, and the middle panel sits inside the literature's standard-error band.

the pathogenic variant is more negative by 2.77 nats, as expected under Equation 1's sign convention. The pair was hand-picked for LLR signal: their delta_norm values (0.030 vs 0.027) sit near the benign median and are visually indistinguishable, while their LLRs separate cleanly. Figure 2 (left panel) shows the $n = 2$ artefact; the middle panel shows the $n = 500$ population context.

Long-protein slice. Among the 236 variants on proteins longer than 1,022 aa, scored via Brandes' variant-centered window, all LLRs are finite (no $|\text{LLR}| > 50$ outliers) and the length-stratified AUROC is within noise of the global figure. The longest protein scored (ASPM, 3,477 aa) yields a finite LLR of -8.20 on a pathogenic variant, consistent with the center-windowed strategy.

5 Discussion

The 500-variant workshop set is the harness's smallest interesting output. The harness's larger output is the loop that produced it: a single natural-language prompt drove dataset construction, encoder selection, scoring, bootstrap, and figure regeneration. Two design choices made that loop tractable.

The prototype is the live demo; the library is the handoff. The notebook and the cluster path share a contract (`ARCHITECTURE.md` §3): they expose compatible encoders, take the same truncation primitive, and produce compatible outputs. The agent picks the right surface for the task it was given. The live demo runs in the notebook's prototype scope on a laptop; the cluster sweep runs in the library scope on SLURM. A new backend — a hosted-inference ESM, a distilled student, an Evo2 [Nguyen et al., 2024] or ESM3 [Lin et al., 2023] drop-in — subclasses one of the two and inherits the rest of the harness.

Methodology bugs the harness caught. Two scars in the week before the workshop, both visible in `experiments/EXPERIMENT_LOG.md`, illustrate why the harness has to be opinionated about the methodology in code rather than in prose.

First, the LLR formula and sign were both wrong (2026-05-13). An earlier implementation of `compute_llr` used a two-pass formula — $\log P_{\mathrm{wt}}(a_{\mathrm{wt}} \mid \mathrm{WT\_seq}) - \log P_{\mathrm{mut}}(a_{\mathrm{mut}} \mid \mathrm{MUT\_seq})$ — with an inverted sign convention claiming "LLR > 0 means pathogenic". AUROC happened to be 0.929 because the two quantities are highly correlated, but the formula and the sign were both wrong. The fix to the Brandes single-pass form (Equation 1) moved AUROC from 0.929 to 0.9300 and put the sign back where Brandes 2023 has it. The validator caught it because the demo pair's pathogenic LLR was no longer more negative than the benign LLR after the sign flip; the smoke test refused to ship.

Second, `MAX_LEN` was off by two (2026-05-12). The constant was set to 1,024, ignoring the BOS and EOS tokens the tokenizer adds. Most variants fit in the 1,022-residue effective window, but the BRCA1 demo pair lives at position 1854 on a 1,863-aa protein — exactly the range where the off-by-two pushes the mutation out of the truncation window. The notebook crashed on the live demo pair before the fix, and the fix shifted the headline AUROC from 0.925 to 0.929 (and then to 0.9300 after the LLR fix above). The harness caught it because the validators run on the demo pair before they run on the 500-variant set; the demo pair was the canary.

Both bugs were caught because the harness has a single source of truth for the formula (`vep_utils.compute_llr`), a single source of truth for the data (`workshop_set_manifest.json`), and a validator that compares the live code against both. Drift in any one of the three surfaces shows up as a validator failure.

6 Limitations

The workshop set is 500 variants, not 36,537. Our 0.025-AUROC headroom over Brandes is consistent with sampling variance, but a confirmatory re-run on the full Brandes benchmark would tighten the comparison; we have not done that re-run. Per-gene AUROC requires at least three pathogenic and three benign rows; 338 of our 400 genes are singletons, so per-gene breakdowns work only for a minority of the panel. Hot genes with label-skewed distributions (FBN1: 11 P / 0 B, LDLR: 6 P / 0 B) are not evaluable on a per-gene basis at all and are excluded from gene-stratified analyses, not from the global AUROC. Label noise from the Conflicting-binarized branch (166 variants, 33%) propagates into the headline; we report the canonical-only AUROC of 0.944 in the provenance log as an upper bound.

The agent itself is the other limitation. The harness reduces a prompt-to-paper run to one natural-language ask, but the agent still makes the wrong call when the prompt is ambiguous; the four-revision history of the workshop set (`experiments/PROVENANCE.md`) catalogues four such wrong calls caught before any slide shipped. Treat the harness as a force multiplier, not a substitute for review.

7 Conclusion

A harness — not a model, not an evaluation — is the load-bearing artefact. Given one, an agent reproduces Brandes 2023 on a 500-variant workshop set at AUROC 0.930 with a single natural-language prompt, and edits this paper to match. Given none, every variant scored requires a human to read a different paragraph of three different manuals first. The harness's commitments — a single source of truth for the methodology, a single source of truth for the dataset, validators that compare the two — are what let the agent run unsupervised.

Acknowledgments

We thank the workshop attendees at Upper Bound 2026 for stress-testing the harness in real time. The submodule pin discipline that kept this reproducible is owed to the `expaper` convention; the substrate-routing dispatcher is shared with the upstream `latent-reasoning-works/shop` skill set.

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